

Real-Time Crypto Fraud Attribution & Rapid Freeze System

****Problem Statement ID:**** 26183 (SIH26183)[cite: 5, 7]

****Category:**** Blockchain Forensics / Cybersecurity[cite: 5]

1. Official Problem Description

Cyber fraud victims increasingly report suspect cryptocurrency wallet addresses used by fraudsters for collection of funds in cases involving investment scams, task-based frauds, sextortion, ransomware, phishing, darknet transactions, and organized cyber-enabled financial crimes[cite: 7].

During investigations, reported wallet addresses are often non-custodial wallets, temporary burner wallets, or intermediary wallets used for layering and laundering[cite: 7]. The inability to quickly identify the cryptocurrency exchange or Virtual Asset Service Provider (VASP) associated with these wallets delays asset freezing, preservation of evidence, tracing of fund flows, and victim fund recovery[cite: 7]. Manual blockchain tracing requires significant technical expertise and time, particularly in cases involving multi-chain transfers, DeFi protocols, mixers/tumblers, bridges, and privacy-enhancing mechanisms[cite: 7].

****Expected Solution:**** A software platform capable of real-time blockchain intelligence generation, automated VASP identification, cross-chain transaction analytics, fund-flow visualization, integration with LEA (Law Enforcement Agency) systems like SAHYOG and NCRP, and generation of standardized investigation reports[cite: 7].

2. Important Concepts (Team Glossary)

- * ****Blockchain & TX Hash:**** A decentralized digital ledger[cite: 4]. A Transaction Hash (TX Hash) is a unique digital receipt proving a specific transfer occurred[cite: 7].
- * ****VASP (Virtual Asset Service Provider):**** Regulated centralized cryptocurrency exchanges (e.g., CoinDCX, Binance)[cite: 7]. These platforms require government IDs (KYC) to open an account[cite: 7]. Identifying the VASP where stolen crypto is deposited allows police to issue legal freeze notices and unmask the criminal[cite: 7].
- * ****Non-Custodial Wallet:**** A private digital wallet where the user holds their own keys (no central company controls it), meaning police cannot freeze it directly[cite: 7].
- * ****Burner Wallet:**** A temporary non-custodial wallet created by fraudsters to receive illicit funds and instantly route them elsewhere before abandoning the wallet[cite: 7].
- * ****Layering & Peel Chains:**** A money-laundering technique where illicit funds are bounced through dozens of intermediary wallets[cite: 7]. A "peel chain" specifically routes a large amount into a wallet,

peels off a small payment to a destination, and sends the bulk remainder to a newly generated change address[cite: 5, 7].

* **Structuring (Smurfing):** Splitting a large illicit sum into dozens of small, identical micro-transactions to evade monitoring thresholds[cite: 5, 7].

* **Mixer / Tumbler:** A privacy smart contract (e.g., Tornado Cash) that pools crypto from hundreds of users, scrambles the coins, and allows withdrawals to fresh addresses to break the tracing link[cite: 7].

* **Blockchain Bridge:** A protocol that moves cryptocurrency from one distinct blockchain (like Ethereum) to another (like Solana), forcing investigators to trace across multiple networks[cite: 7].

* **NCRP / 14C / SAHYOG:** The NCRP is India's National Cybercrime Reporting Portal where victims report fraud[cite: 5, 7]. The 14C (Indian Cyber Crime Coordination Centre) coordinates the suspect registry, and SAHYOG is the unified government portal used to issue fast-tracked legal freeze notices to VASPs[cite: 5, 7].

* **Section 63 BSA:** The Bharatiya Sakshya Adhiniyam provision that governs electronic evidence admissibility in Indian courts, requiring digital certificates and cryptographic hashes to prove data wasn't tampered with[cite: 5].

3. End-to-End System Workflow

Stage 1: Complaint Ingestion & Evidence Locking

1. **[Victim Submits Complaint]**[cite: 3]
2. **[Format & Checksum Validation]** → Rejects invalid addresses immediately; if Pass, proceeds[cite: 3].
3. **[On-Chain Verification (RPC/Explorer API)]** → Checks if TX Hash actually exists on-chain and confirms Sender, Receiver, Amount & Timestamp match claim[cite: 3].
4. **[Extract Seed Target Address]** → Parse confirmed transaction output to isolate the scammer's receiving wallet (Hop 0 Target) for outbound tracing[cite: 3].
5. **[Chain-of-Custody Hashing & Audit Log]** → Creates Intake Package: (NCRP Ack No + UTC Timestamp + Victim Address + TX Hash + Seed Target Address). Generates an immutable SHA-256 Hash and stores it in an Audit Ledger[cite: 3].
6. **[Cross-Case Correlation Lookup]** → Searches internal DB for matching Suspect Address/Cluster. Flags as [SERIAL FRAUD CLUSTER] or [STANDARD LEAD][cite: 3].

Stage 2: Tracing & Graph Expansion Engine

1. **[Start: Seed Target Address (Hop 0)]**^[cite: 3]
2. **[Outbound Transaction Scanner]** → Applies time filter ($\geq$ Fraud Timestamp)^[cite: 3].
3. **[Node Classification & Graph Safeguards]**
 - * Is Mixer/Tumbler? → Flags as Terminal Dead-End & Exports Pre-Mixer Trace^[cite: 3].
 - * Is High-Degree (>500 txs) / VASP Hub? → Triggers Early Exit to stop BFS Fan-Out^[cite: 3].
 - * Is Cross-Chain Bridge / DEX? → Applies Probabilistic Amount/Time Correlation Window^[cite: 3].
4. **[Laundering Heuristic Engine & Risk Scoring]** → Checks for Peel Chains (Asymmetric Split $>80\%$ Change / $<20\%$ Peel), Structuring (rapid fan-out), and Fund Velocity (<24 h holding time) to calculate a Risk Score (1.0 - 10.0)^[cite: 3].
5. **[Hop Depth Check]**
 - * Depth < 8 & Non-Terminal → Pushes Next Hop to BFS Queue (Loop Scanner)^[cite: 3].
 - * Terminal Reached / Depth ≥ 8 → Locks Path & Extracts Terminal Entity^[cite: 3].
6. **[VASP Classification & Confidence Gating]**
 - * Domestic VASP (Confidence $\geq 80\%$) → Auto-Drafts SAHYOG / I4C Freeze Order^[cite: 3].
 - * Domestic VASP (Confidence $< 80\%$) → Flags as Investigative Lead for Manual Review^[cite: 3].
 - * Offshore / Unregulated VASP → Auto-Drafts FIU-IND / Interpol Referral^[cite: 3].

Stage 3: Evidentiary Reporting & Visual Export

1. **[Generate Standardized Evidentiary JSON Payload]** → Structures Nodes (Hop No., Risk Score, Entity Tag, Tx Count, Balance) and Edges (TX Hash, Amount, Timestamp, Heuristic Reason)^[cite: 3].
2. **[Cryptographic Sealing & Legal Certification]** → Calculates a final SHA-256 Hash of the JSON Payload, logs it in the Audit Ledger, and Auto-Generates a Statutory Section 63 BSA Digital Evidence Certificate^[cite: 3].
3. **[Interactive Visual Dashboard Rendering]** → Renders an Interactive Force-Directed Graph using D3.js/React Flow with Color-Coded Risk nodes (Green=Safe, Amber=Hop, Red=Fraud, Blue=VASP). An Interactive Sidebar allows investigators to click nodes to inspect on-chain metrics without raw key access^[cite: 3].
4. **[One-Click Export]** → Packages the Structured JSON, Signed PDF Legal Notice, and BSA Certificate^[cite: 3].

4. Critical Flaws Solved (Judge's Review Matrix)

1. ****No Confidence Gate (CRITICAL):**** Fixed by enforcing an 80% confidence threshold before auto-drafting a legal freeze notice, preventing the system from freezing innocent exchange deposit accounts[cite: 3, 5, 6].
2. ****In-Memory Graph Explosion (HIGH):**** Fixed by capping Breadth-First Search (BFS) depth to < 8 hops and hard-stopping the trace on high-degree nodes (>500 transactions/day) to prevent memory crashes[cite: 3, 5, 6].
3. ****VASP Label Sourcing (HIGH):**** Fixed by scoping the initial database to curated Indian VASPs (CoinDCX, WazirX) and publicly sanctioned OFAC addresses, with a roadmap for API data-sharing MoUs[cite: 5, 6].
4. ****No Chain-of-Custody (CRITICAL):**** Fixed by applying SHA-256 hash timestamping at ingestion and output, and binding the trace to a Section 63 BSA certificate for court admissibility[cite: 3, 5, 6].
5. ****Deterministic Bridge Tracing (MEDIUM):**** Fixed by treating blockchain bridges probabilistically (correlating amount minus fees within a time window) rather than as guaranteed deterministic links[cite: 3, 5].
6. ****Offshore VASP Dead-End (HIGH):**** Fixed by auto-routing terminal traces at offshore/non-compliant VASPs to an FIU-IND / Interpol referral template instead of a SAHYOG notice[cite: 3, 5, 6].
7. ****Address Input Validation (LOW):**** Fixed by validating Base58Check (BTC) and EIP-55 (Ethereum/Tron) checksum formats immediately upon victim submission[cite: 3, 5].

5. Software Architecture & Resources Required

Backend Engine & Logic Layer

- * ****Language:**** Python 3.11 with FastAPI (for asynchronous blockchain indexing without blocking memory).
- * ****Graph Logic:**** Python `NetworkX` library (used to map BFS logic, calculate peel chain asymmetry, and identify structuring heuristics).
- * ****PDF/Evidence Generation:**** Python `ReportLab` (for automating the SAHYOG notice PDF and Section 63 BSA legal certificate).

Database Layer

- * ****Graph Database:**** Neo4j (AuraDB). Required for storing nodes and edges efficiently and preventing server crashes on multi-hop tracing[cite: 5].
- * ****Relational Database / Audit Ledger:**** PostgreSQL (Supabase). Used strictly as an append-only ledger to store NCRP case IDs, timestamps, and SHA-256 cryptographic hashes[cite: 3].

Frontend Dashboard

- * **Framework:** React (Next.js) styled with Tailwind CSS.
- * **Visual Renderer:** `@xyflow/react` (React Flow) or Cytoscape.js. Enables the floating, color-coded interactive graph where users can click nodes to view sidebars containing transaction metrics without leaving the platform[cite: 3].

External APIs & Datasets (Data Sourcing)

- * **TronGrid API** (`trongrid.io`): Crucial for fetching TRC-20 USDT transaction data (the most common fraud vector). Free tier available.
- * **Etherscan & Alchemy APIs** (`etherscan.io`): For tracking Ethereum transactions, ERC-20 tokens, and smart contract (Mixer/Bridge) interactions. Free tier available.
- * **Blockstream.info API** (`blockstream.info/api`): To trace Bitcoin UTXO flows (inputs/outputs for co-spend clustering)[cite: 4].
- * **OFAC SDN Crypto Blacklist:** Open-source datasets (e.g., GitHub repositories tracking US Treasury sanctions) to flag known terrorist and laundering wallets[cite: 5].
- * **CryptoScamDB:** Open-source blacklist to identify malicious smart contracts, phishing drainers, and mixers.